

# Improving Rare-Outcome Prediction With Separability-Aware Synthetic Health Data: A Four-Task Validation Study

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## Abstract

**Background:** Synthetic data are often used to address rare outcomes by increasing minority-class sample size, but additional records may not improve prediction when classes overlap or when generation fails to preserve useful class-conditional structure.

**Objective:** This study asks when controlling class separability improves the held-out-real utility of synthetic health data beyond class-frequency correction and whether utility gains are accompanied by loss of minority coverage.

**Methods:** We developed SEPAWARE, a post-generation mechanism that selects records from an oversized candidate pool using realism and label-consistent geometric evidence while preserving a prespecified class count. Validation uses four outcome tasks from three health-data sources: Vigitel obesity, COVID-19 hospital mortality, Vigitel current smoking, and landmark-defined 365-day kidney-cohort mortality. Completed analyses include leakage-controlled 5-fold synthetic-to-real evaluation and a blocked  $2 \times 2$  factorial mechanism experiment. A strengthened augmentation analysis uses three repeated 5-fold partitions and compares real-only training, random sampling, SMOTE, class weighting, conditional tabular generative adversarial networks, tabular variational autoencoders, and their SEPAWARE-selected variants. Evaluation includes discrimination, minority-specific performance, calibration, class-conditional coverage, and empirical privacy-risk proxies. A controlled simulation independently varies prevalence and overlap.

**Results:** In the synthetic-to-real experiment, Vigitel Macro-F1 increased from 0.459 with standard conditional tabular generative adversarial network data to 0.583 with SEPAWARE, and separability accounted for 96.00% of treatment-plus-error variation ( $P < .001$ ). In the strengthened analysis, SEPAWARE improved mean AUPRC over standard CTGAN by 0.015 on COVID-19, 0.013 on Vigitel obesity, and 0.037 on Vigitel smoking, but by only 0.0003 on kidney mortality. It was inconsistent for TVAE and did not generally exceed real-only training or simple rebalancing. Selection improved nearest-minority coverage while reducing diversity, including for kidney mortality where no meaningful utility gain appeared. Controlled SepAware-SMOTE gains were small and its overlap and prevalence interactions were not significant.

**Conclusions:** Class-boundary control is distinct from class balancing and can improve some candidate pools, but it is not universally beneficial. The kidney-mortality null result and repeated diversity losses show that apparent gains must be tested against simple baselines, calibration, minority coverage, and the possibility that selection merely removes difficult but valid cases.

**Keywords:** synthetic health data; tabular data; class imbalance; class overlap; rare outcomes; data augmentation; privacy; calibration

## 1 Introduction

Health datasets combine sensitive information, mixed numerical and categorical variables, missingness, and outcomes that are often uncommon but clinically consequential. Synthetic tabular data

can support model development and collaboration when access to real records is restricted, but generation alone does not guarantee privacy or downstream usefulness [1–3]. In particular, a generator may preserve aggregate distributions while weakening the class-conditional structure needed to identify a rare outcome or may yield deceptively easy synthetic examples [4, 5].

Rare-outcome learning is commonly framed as a class-imbalance problem. Random resampling, the Synthetic Minority Over-sampling Technique (SMOTE), class weighting, and conditional generation increase the number or influence of positive-class observations [6, 7]. More data, however, need not provide more information when positive and negative observations overlap in feature space. Class prevalence and class overlap are distinct properties: data-level or algorithm-level imbalance correction changes representation or decision cost, whereas it does not necessarily repair weak class-conditional geometry [7].

Existing synthetic-health-data evaluations typically assess global resemblance, downstream utility, and privacy after generation [1, 2, 5]. They provide less guidance on whether post-generation control of class geometry improves rare-outcome prediction, when such control fails, and whether apparent gains arise by discarding difficult but legitimate positive cases. This gap motivates the present study. Our objective is to determine whether separability-aware candidate selection adds predictive value beyond class-count correction and standard generation, while explicitly measuring calibration, positive-class coverage and diversity, and empirical privacy risk.

We developed SEPAWARE, a post-generation mechanism that draws an oversized candidate pool and ranks records using proximity to the real training manifold and label-consistent geometric evidence. It retains a prespecified class count, separating the effect of candidate selection from a change in class ratio. We first isolate the mechanism in Vigitel obesity and COVID-19 hospital mortality, then evaluate practical augmentation across those tasks plus Vigitel current smoking and landmark-defined kidney-cohort mortality.

The principal result is deliberately conditional. Within CTGAN candidate pools, SEPAWARE increased mean AUPRC relative to standard CTGAN by 0.015 for COVID-19 mortality, 0.013 for Vigitel obesity, and 0.037 for Vigitel smoking, but by only 0.0003 for kidney mortality. Real-only training or simple resampling remained competitive or superior on several endpoints, TVAE results were inconsistent, and improved positive-class coverage repeatedly coincided with reduced diversity. Thus separability control can help in some settings, but it is neither a universal substitute for imbalance correction nor evidence that selected records preserve all clinically meaningful heterogeneity.

## 1.1 Research Questions

The study addresses three research questions:

- RQ1:** Is data-level or algorithm-level class-imbalance correction sufficient, or does class overlap limit the utility of synthetic augmentation?
- RQ2:** Does separability-aware selection improve rare-outcome prediction beyond random sampling, SMOTE, class weighting, and unconstrained generation?
- RQ3:** When predictive performance improves, is the gain accompanied by acceptable calibration, positive-class coverage and diversity, and empirical privacy risk?

The mechanism experiment uses leakage-controlled synthetic-to-real evaluation followed by a blocked  $2 \times 2$  factorial analysis that separates separability weighting from clinical-anchor weighting. The strengthened experiment evaluates real-plus-synthetic augmentation on shared held-out real folds using real-only training, conventional imbalance methods, CTGAN, TVAE, and SEPAWARE

variants. AUPRC is the primary augmentation endpoint; secondary analyses include Macro-F1, positive-class metrics, AUROC, balanced accuracy, Brier score, calibration, coverage, diversity, and privacy-risk proxies.

## 1.2 Contributions

This four-task study makes four contributions:

1. It frames class overlap as a target distinct from class imbalance and global fidelity in synthetic health-data augmentation.
2. It introduces an auditable post-generation selection mechanism whose effect can be separated from generator fitting and class-count correction.
3. It uses a factorial experiment to attribute the strong Vigitel obesity synthetic-to-real change to separability weighting while reporting the COVID-19 null attribution with equal emphasis.
4. It tests the easier-case explanation through a strengthened benchmark spanning four tasks, two generator families, conventional imbalance baselines, calibration, positive-class structure, and privacy diagnostics.

The remainder of the paper reviews synthetic tabular health data and rare-outcome evaluation, defines the datasets and leakage-controlled protocols, reports the mechanism and four-task augmentation results, interprets their boundary conditions, and concludes with limitations and requirements for external validation.

## 2 Related Work

This section positions SEPAWARE within four connected areas: synthesis of mixed-type health tables, evaluation beyond aggregate resemblance, rare-outcome learning under imbalance and overlap, and the privacy–utility tension. Each area supplies part of the required evaluation, but their intersection remains underexplored.

### 2.1 Synthetic Tabular Health Data

Tabular health data represent one person or encounter per row and heterogeneous clinical or survey variables in columns; synthetic microdata are artificial row-level records intended to reproduce selected properties of such data without directly reproducing the source rows. Statistical microdata synthesis predates deep learning. Multiple imputation and probabilistic graphical models generate records from estimated conditional distributions and remain attractive because their assumptions can be inspected [8]. Copulas provide efficient dependence modelling but can struggle when discrete, continuous, multimodal, and rare-event variables interact nonlinearly [9].

Generative adversarial networks formulate generation as an adversarial game [10]. MedGAN adapted this approach to high-dimensional discrete patient records [11], and CTGAN addressed mixed-type tables through mode-specific normalization, conditional vectors, and training-by-sampling [12]. Variational autoencoders optimize a likelihood-related lower bound [13]; TVAE is a tabular extension evaluated alongside CTGAN [12]. TabDDPM later applied separate diffusion processes to numerical and categorical attributes and outperformed several GAN and VAE comparators on general tabular benchmarks [14]. These methods improve generation, whereas SEPAWARE operates after generation: it tests whether transparent control of the candidate pool’s

class geometry can improve rare-outcome utility. We use CTGAN and TVAE to examine transfer across two generator families; diffusion remains a future comparator.

## 2.2 Evaluation Beyond Aggregate Resemblance

Health-synthesis studies repeatedly conclude that no single metric establishes usefulness or safety and that evaluation should jointly cover resemblance, downstream utility, and privacy [1, 2, 5]. Recent comparisons report dataset- and metric-dependent rankings and a persistent fidelity/utility–privacy trade-off [15, 16]. A review of structured synthetic electronic health records also documents continuing problems with rare events, reproducibility, and privacy evaluation [4].

Univariate fidelity measures whether synthetic columns resemble their real counterparts, pairwise fidelity evaluates dependence structure, and multivariate distinguishability tests whether a classifier separates real from synthetic records [1, 2]. None alone establishes rare-outcome usefulness: aggregate scores can be dominated by negative-class records, whereas a selector can improve classification by producing easy positive prototypes rather than a faithful positive population [4, 5].

Train-on-Synthetic, Test-on-Real (TSTR) evaluation probes whether relations learned from synthetic records transfer to held-out real observations; Real-plus-Synthetic-to-Real evaluation instead tests augmentation [1, 2]. These regimes answer different questions and are reported separately here. Neither constitutes a privacy guarantee or proves unbiased clinical inference.

## 2.3 Class Imbalance, Overlap, and Augmentation

Accuracy can remain high while positive-class recall is poor in imbalanced data [7]. Macro-F1 gives both classes equal weight, while AUPRC is often more informative than AUROC for rare positive outcomes [17]. Conventional approaches include random resampling, SMOTE [6], cost-sensitive learning, and threshold adjustment [7]. These methods alter frequency or decision cost but do not necessarily recover informative class-conditional structure.

A recent comparison across 13 small health datasets found that sampling with replacement could match more complex generative augmentation in several settings, suggesting that some apparent gains arise mainly from increased sample size [18]. This makes random oversampling a necessary comparator and sharpens our question: does controlling generated class geometry add value beyond correcting sample counts?

Class imbalance and class overlap are related but distinct [7]. A rare positive class can remain geometrically well separated, whereas a less imbalanced outcome can be intrinsically difficult because valid observations occupy overlapping feature regions. Adding examples cannot remove irreducible overlap, and aggressively selecting distant examples may improve an empirical boundary by deleting difficult but legitimate cases. We therefore interpret discrimination jointly with calibration and positive-class coverage and diversity.

## 2.4 Privacy–Utility Tension

Synthetic data are not anonymous by construction: high-capacity generators can reproduce or reveal information about training records [3, 19]. Privacy can be examined through nearest-record, membership-inference, and attribute-inference analyses or protected through formal mechanisms such as differential privacy [20]. LDP-GAN combines local differential privacy with medical-record synthesis [21], while a recent oncology framework evaluates candidate synthetic datasets across multiple quality dimensions before downstream use [22]. Accordingly, our study reports empirical privacy proxies separately from fidelity and utility and does not equate a near-chance proxy with anonymity.

## 2.5 Research Gap and Study Positioning

Table 1 summarizes how representative studies address the dimensions central to this work. The literature leaves a practical gap at the intersection of rare outcomes, explicit class-overlap control, held-out-real augmentation, positive-class coverage, and privacy diagnostics. Generators optimize distribution learning, imbalance methods correct representation or decision cost, and evaluation frameworks generally assess outputs after generation rather than steering class-conditional geometry. Our contribution is not a universally superior generator; it is an auditable intervention and evaluation design for determining when separability-aware selection helps and what it may sacrifice.

Table 1: Positioning of representative related work. A checkmark denotes an explicit focus; a dash denotes that the dimension is not the study’s primary contribution.

| Study                      | Health | Rare<br>outcome | Overlap | Utility | Privacy |
|----------------------------|--------|-----------------|---------|---------|---------|
| MedGAN [11]                | ✓      | –               | –       | ✓       | –       |
| CTGAN/TVAE [12]            | –      | –               | –       | ✓       | –       |
| Health-data evaluation [1] | ✓      | –               | –       | ✓       | ✓       |
| Structured-EHR review [4]  | ✓      | ✓               | –       | ✓       | ✓       |
| LDP-GAN [21]               | ✓      | –               | –       | ✓       | ✓       |
| This study                 | ✓      | ✓               | ✓       | ✓       | ✓       |

## 3 Materials and Methods

This section defines the four outcome tasks, leakage-controlled evaluation regimes, generators, selection rule, factorial mechanism analysis, strengthened augmentation benchmark, controlled overlap experiment, and reproducibility safeguards.

### 3.1 Datasets Definition and Description

#### 3.1.1 Vigitel health survey

Vigitel is Brazil’s telephone-based surveillance system for risk and protective factors for chronic non-communicable diseases [23]. The obesity experiments use a reproducible stratified sample of 5,000 rows drawn from a 2006–2023 extract, comprising 4,149 non-obese and 851 obese records (imbalance ratio approximately 4.9:1). The factorial experiment uses 15 predictors: age, sex, current and former smoking, physical inactivity, leisure-time physical activity, diabetes, hypertension, dyslipidemia, dietary indicators, fruit and vegetable regularity, and schooling. Not every variable in the source extract is used; identifiers, the outcome, and variables excluded by the prespecified analytic pipeline are omitted.

#### 3.1.2 COVID-19 hospital mortality

The COVID-19 dataset contains 334 hospital records and 34 columns, including the binary target `obito` (death). The executed experiments report 251 survivors and 83 deaths, an imbalance ratio of approximately 3.0:1. Predictors include age, sex, alcohol and tobacco indicators, admission-reason groups, number of comorbidities, and specific conditions such as hypertension, heart failure, coronary disease, chronic respiratory disease, diabetes, chronic kidney disease, cancer, cirrhosis, and HIV.

The small sample size and the clinical heterogeneity of the death class are scientifically important rather than incidental. A synthetic-selection method can appear to improve separability simply by concentrating on an artificially homogeneous subset of deaths; results on this dataset must therefore be interpreted with wide uncertainty and assessed strictly on held-out real folds.

### 3.1.3 Vigitel current smoking

The third task uses the supplied 21,764-record Vigitel analytic sample with current smoking (`fumante`) as the outcome. The executed experiment uses a reproducible stratified sample of 5,000 records: 616 current smokers and 4,384 nonsmokers. Identifiers, colour labels, and upstream cluster fields are excluded; 27 demographic, socioeconomic, diet, activity, and chronic-condition variables are retained. Because the supplied file has no survey-year or sampling-weight field, results describe the analytic sample rather than a weighted population parameter.

### 3.1.4 Kidney-cohort 365-day mortality

The fourth task is derived from 9,559,286 dated events for 67,267 patients. Each patient’s landmark is 365 days after first recorded activity, and 15 predictor counts are computed strictly before that date. The outcome is recorded death during the subsequent 365 days. Patients who died before the landmark are excluded; nondeaths require observable activity through the horizon. The eligible cohort has a 5.34% event rate; the executed stratified sample contains 5,000 patients, 267 deaths and 4,733 nondeaths. Dialysis is not an outcome in this study.

Table 2 summarizes the sample sizes, class counts, and experimental roles of all four tasks.

Table 2: Datasets used in the reported experiments. Counts are taken from executed repository outputs.

| Dataset                  | $n$   | Positive | Negative | Primary use                                     |
|--------------------------|-------|----------|----------|-------------------------------------------------|
| Vigitel obesity sample   | 5,000 | 851      | 4,149    | SEPAWARE mechanism and augmentation experiments |
| COVID-19 mortality       | 334   | 83       | 251      | SEPAWARE mechanism and augmentation experiments |
| Vigitel smoking sample   | 5,000 | 616      | 4,384    | Strengthened augmentation validation            |
| Kidney 365-day mortality | 5,000 | 267      | 4,733    | Landmark augmentation validation                |

## 3.2 Preprocessing and Cross-validation Protocol

All predictive experiments use stratified five-fold cross-validation. Binary variables are normalized to  $\{0, 1\}$ , and 1/2-coded survey fields are mapped deterministically. Within each fold, numerical missing values are median-imputed, binary fields use the modal category, and continuous variables are standardized to zero mean and unit variance using statistics fitted only on that fold’s real training subset. CTGAN fitting, candidate scoring, selection, and classifier training all occur within the training portion; the held-out real fold is used only for evaluation.

Every predictive result reported in Section 4 follows this fold-contained protocol. Within each fold of the factorial experiment, a single CTGAN model generates a candidate pool five times the real training size, and the same pool is scored under all four selection conditions described in Section 3.8. Pairing removes generator-to-generator variation between conditions, but the five fold scores are not independent replications: training folds overlap by 75%, and the four conditions

within a fold share a candidate pool. Section 5.6 discusses this dependence, and the primary factorial model blocks on fold.

We distinguish exploratory from confirmatory evidence throughout. All numerical claims in Sections 4 and 5 come from executed analyses and exported result tables; protocol-only, superseded single-split, or failed runs are not reported as evidence (Appendix A). Results from different evaluation regimes are not pooled because standalone synthetic substitution, real-plus-synthetic augmentation, and internal synthetic cross-validation answer different questions.

### 3.3 Evaluation Regimes

Three regimes are kept distinct throughout:

**Real→Real (TRTR):**

*train on four real folds and evaluate on the fifth real fold, this is the predictive reference.*

**Synthetic→Real (TSTR):**

*train only on synthetic records generated from a real training fold and test on the corresponding held-out real fold, this measures substitution utility.*

**Real+Synthetic→Real:**

*append synthetic minority records to the real training fold and test on held-out real data, this measures augmentation, not synthetic substitution.*

Macro-F1 is the endpoint of the original mechanism experiments because both outcome classes contribute equally. In the strengthened augmentation analysis, AUPRC is the prespecified primary endpoint. Secondary outcomes are Macro-F1, positive-class F1, precision and recall, AUROC, balanced accuracy, Brier score, and calibration intercept and slope. Metrics not produced by the original executed pipeline are reported as unavailable rather than retrospectively reconstructed.

### 3.4 Generators and Predictive Classifiers

CTGAN [12] is the generator in the mechanism experiments. The strengthened augmentation analysis adds TVAE [12] to test whether the selection mechanism transfers beyond a GAN. Standard-generator and SEPARE conditions share a candidate pool within a fold. Because Synthetic Data Vault (SDV) version 1.17 does not implement native conditional sampling for TVAE, TVAE is fitted directly to positive-class training-fold feature records and the outcome label is assigned by design; this is reported separately from CTGAN’s outcome-conditional sampling. Predictive models in the strengthened analysis are logistic regression, random forest, and CatBoost [24]; the mechanism experiments also use XGBoost [25] and TabPFN [26].

### 3.5 SepAware Selection Framework

For real training data  $\mathcal{D}_R$  and candidate pool  $\mathcal{C}$ , each candidate  $(\tilde{x}, \tilde{y})$  receives a composite score

$$S(\tilde{x}, \tilde{y}) = R(\tilde{x}) + \lambda_{\text{sep}} \text{Sep}(\tilde{x}, \tilde{y}) + \lambda_{\text{anchor}} A(\tilde{x}, \tilde{y}), \quad (1)$$

where  $R$  is one minus a robustly normalized nearest-real distance,  $\text{Sep}$  averages a nearest-hit/nearest-miss margin and the assigned-label probability from a real-trained random forest, and  $A$  averages compatibility with predeclared class-conditional anchor patterns. The nonnegative

weights  $\lambda_{\text{sep}}$  and  $\lambda_{\text{anchor}}$  control the contributions of separability and anchor compatibility, respectively. Candidates are ranked within target class and selected to reproduce the real training-fold class counts, so improved TSTR performance cannot be attributed to a changed class ratio.

The fixed- $\lambda$  grid uses  $\lambda_{\text{sep}} \in \{0, 0.25, 0.5, 1\}$  and  $\lambda_{\text{anchor}} \in \{0, 0.25, 0.5\}$ , because the best configuration on this grid is chosen using the same outcomes later used for reporting, grid results are treated as exploratory (Section 4.2). The subsequent factorial experiment instead predeclares two levels per factor,  $\lambda_{\text{sep}} \in \{0, 1\}$  and  $\lambda_{\text{anchor}} \in \{0, 0.5\}$ , so that its conclusions do not depend on post-hoc selection.

Figure 1 summarizes the pipeline. CTGAN produces candidates but does not itself optimize the three selection criteria; a separate selector scores and ranks candidates within each class and returns the requested class counts. Separating generation from selection makes the intervention auditable and allows component-wise ablation, which is the basis of the factorial design in Section 3.8. The design rules out one simple alternative explanation – that a TSTR change is caused by a different class ratio – because class counts are held fixed across all compared conditions. It does not rule out harder alternatives: ranking can select unusually easy prototypes, discard boundary cases, or amplify a classifier-specific shortcut. These possibilities motivate the confirmatory analyses proposed in Section 6.

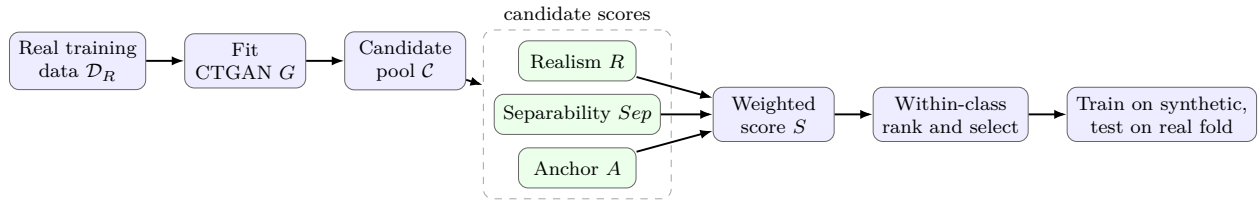


Figure 1: SEPAWARE post-generation selection pipeline: every operation is fitted or computed using the real training portion of a cross-validation fold; the held-out real fold is used only for evaluation.

### 3.6 Fidelity, Separability, and Privacy Proxies

Fidelity combines numerical-distribution similarity, categorical-proportion similarity, and correlation preservation into a single composite score. Separability is reported using cosine silhouette, a one-nearest-neighbor label-agreement index (SI), and hypothesis margin (HM). These are descriptive geometric measures: higher separation is not automatically clinically desirable, and we do not treat it as such (Section 5).

The strengthened experiment measures held-out-real positive-class coverage as the mean distance from each positive test record to its nearest synthetic positive record after training-fold standardization. Positive-class diversity is the mean pairwise standardized distance among up to 1000 selected positive records. Exact overlap is the proportion of synthetic records with zero distance to a real training record. The membership-inference proxy compares each real training and held-out record’s distance to the nearest synthetic record and reports membership-classification AUROC. These are empirical diagnostics, not evidence of anonymity or a formal guarantee.

### 3.7 Domain-anchored Causal Plausibility Filtering

DACPF (Domain-Anchored Causal Plausibility Filtering) generates positive-class CTGAN candidates and ranks their consistency with real positive-class distributions for domain anchors. The completed COVID-19 experiment compares no augmentation, standard CTGAN augmentation, age

filtering, disease-anchor filtering, and combined filtering, evaluated as real-plus-synthetic augmentation (Real+Synthetic→Real) rather than standalone synthetic substitution. We emphasize that DACPF is a plausibility filter, not causal discovery or effect estimation: “causal” describes the clinical origin of the anchor hypotheses, not a claim that the filtering step identifies a causal effect.

### 3.8 Blocked $2^2$ Factorial Experiment

The mechanism study uses a two-factor, two-level factorial design: separability weighting is factor  $A$ , clinical-anchor weighting is factor  $B$ , and all four combinations are evaluated on the same fold-specific candidate pool. Fold is included as a block because conditions within a fold are paired rather than independent replications.

Let  $A \in \{-1, +1\}$  encode separability weighting and  $B \in \{-1, +1\}$  anchor weighting. The fold-level response is the classifier-averaged TSTR Macro-F1,

$$y_{ijk} = \mu + q_A A_i + q_B B_j + q_{AB} A_i B_j + \gamma_k + \varepsilon_{ijk}, \quad (2)$$

where  $y_{ijk}$  is the classifier-averaged Macro-F1 for separability level  $i$ , anchor level  $j$ , and fold  $k$ ;  $\mu$  is the grand mean;  $A_i$  and  $B_j$  are the coded factor levels;  $q_A$ ,  $q_B$ , and  $q_{AB}$  are the separability, anchor, and interaction coefficients;  $\gamma_k$  is the fixed effect of fold  $k$ ; and  $\varepsilon_{ijk}$  is residual error. The high-minus-low factorial effects are  $2q_A$ ,  $2q_B$ , and  $2q_{AB}$ . Type-II ANOVA partitions sums of squares among factors, interaction, fold, and residual error; the primary variation percentages exclude the fold block from the denominator so that factor contributions are compared against experimental error rather than against total variation. An unblocked  $2^{2r}$  decomposition that treats folds as replications is also reported in the supplementary material for comparison, but Equation 2 is the more defensible model for paired cross-validation conditions and is used for all primary conclusions.

Both raw and log-transformed responses are analysed. Residual normality is assessed using Shapiro–Wilk and Jarque–Bera tests [27, 28]; constant variance is assessed with median-centred Levene (Brown–Forsythe), Fligner–Killeen, and Breusch–Pagan tests [29–31]. Durbin–Watson and lag-one Ljung–Box statistics diagnose ordering-related serial patterns but cannot establish independence [32, 33]. As noted in Section 3.2, five-fold training sets overlap by 75% and the four conditions within a fold are paired, so inference from this design is treated as exploratory rather than as a formal, independently replicated hypothesis test (Section 5.6).

### 3.9 Strengthened Augmentation Experiment

The augmentation experiment uses three repeated five-fold stratified partitions and compares real-only training, random oversampling, random undersampling, SMOTE, class-weighted training, standard CTGAN, SEPAWARE CTGAN, standard TVAE, and SEPAWARE TVAE. Synthetic conditions add positive candidates until the real training fold is balanced, the held-out fold retains its natural prevalence. All missing-value imputation is fitted on the real training fold before resampling or generation. The SEPAWARE augmentation score uses realism plus separability with a fixed unit separability weight, no held-out outcome is used to tune the setting.

Logistic regression supplies a linear and calibration-sensitive reference, random forest supplies a nonlinear ensemble, and CatBoost supplies a strong tabular learner. Hyperparameters and the 0.5 classification threshold are held fixed across methods. Fold-level results are retained so generator comparisons can be paired within dataset, repeated partition, fold, and classifier. The manuscript uses three independently shuffled five-fold partitions with independent fold-specific generator seeds. These repetitions improve stability assessment but remain internal repeats of the same source samples rather than external validation.

### 3.10 Four-task Strengthened Augmentation Experiment

The strengthened benchmark uses three repeated five-fold stratified partitions for each task. Within each training fold it compares real-only training, class weighting, random over- and undersampling, SMOTE, standard CTGAN, SEPAWARE CTGAN, standard TVAE, and SEPAWARE TVAE. Synthetic conditions add positive candidates until the training fold is balanced; the held-out real fold retains its natural prevalence. CTGAN and TVAE are trained for 30 epochs with fold-specific seeds and candidate pools five times the required positive count. Standard and selected conditions share the same generator fit. Logistic regression, random forest, and CatBoost are evaluated on identical held-out patients. All preprocessing, generation, and selection are fitted within the training fold.

### 3.11 Controlled Prevalence-by-overlap Experiment

A separate mechanism experiment varies class prevalence and overlap independently and is not counted as a third health dataset. We generated 2000-record, 15-feature binary-classification problems with 8 informative features, 3 redundant features, 2 clusters per class, and 2% label noise. Prevalence was set to 5%, 15%, or 30%, and the class separation to 0.4, 1.0, or 2.0, corresponding respectively to high, moderate, and low overlap. Each of the 9 cells used 10 independent data seeds and a stratified 70/30 train-test split.

The comparison includes real-only training, random oversampling, SMOTE, and a SepAware-SMOTE condition. For the latter, SMOTE generates five times the number of required positive-class candidates, the selector ranks them by nearest-real realism, nearest-hit/nearest-miss margin, and random-forest label confidence, then selects enough records to balance the training data. Logistic regression and random forest are evaluated with AUPRC, Macro-F1, and positive-class F1. A heteroscedasticity-robust linear model tests method-by-overlap and method-by-prevalence interactions.

### 3.12 Ethical considerations

The Vigitel analysis uses a secondary survey extract and the COVID-19 analysis uses a de-identified institutional dataset supplied for the qualification research. The manuscript does not infer a new ethics approval number or consent determination. The authors must insert the applicable institutional review board or ethics committee name, protocol identifier, consent or waiver determination, and any access restrictions before submission.

### 3.13 Reproducibility

The strengthened experiment was executed with Python 3.12, NumPy 1.26.4, pandas 2.2.2, scikit-learn 1.6.1, SDV 1.17.2, CatBoost 1.2.10, XGBoost 3.2.0, imbalanced-learn 0.12.3, and statsmodels 0.14.2. A project-local environment pins NumPy 1.26.4 because the available compiled scientific extensions were incompatible with NumPy 2. Full software-stack notes, executed-notebook identifiers, and artefact-status decisions are given in the code and data availability statement and in Appendix [A](#).

## 4 Results

### 4.1 Rebalancing and unconstrained augmentation are insufficient

The project began from the conventional explanation for poor rare-outcome prediction: that the minority class simply supplies too few examples. We first examined ordinary rebalancing, CTGAN minority augmentation, and increasingly targeted candidate filters. The common result across these attempts was not that augmentation never helped, but that additional minority records did not produce a sufficiently large and consistent gain to explain the problem by class frequency alone. Some of these early notebooks were single-split or diagnostic studies and are treated as hypothesis-generating rather than confirmatory evidence.

The leakage-controlled DACPF rerun provides the strongest five-fold evidence from this stage (Table 3). Standard CTGAN augmentation improves mean Macro-F1 relative to real-only training for both classifiers available in the local environment. Adding an age filter, or a combined age-and-disease filter, does not improve on standard CTGAN. Disease-anchor filtering performs best, adding 0.0508 over standard CTGAN for CatBoost but only 0.0084 for XGBoost. Domain targeting can therefore help, but its effect depends on both the anchor and the classifier, and adding more constraints is not monotonically beneficial.

Table 3: Five-fold COVID-19 augmentation Macro-F1. These are Real+Synthetic→Real results and are not directly comparable with the standalone-synthetic TSTR results in Section 4.2.

| Condition                     | CatBoost      | XGBoost       |
|-------------------------------|---------------|---------------|
| Real only                     | 0.4740        | 0.4759        |
| Real + standard CTGAN         | 0.5262        | 0.5255        |
| Real + age-filtered CTGAN     | 0.5257        | 0.4983        |
| Real + disease-filtered CTGAN | <b>0.5771</b> | <b>0.5339</b> |
| Real + combined DACPF         | 0.5256        | 0.5204        |

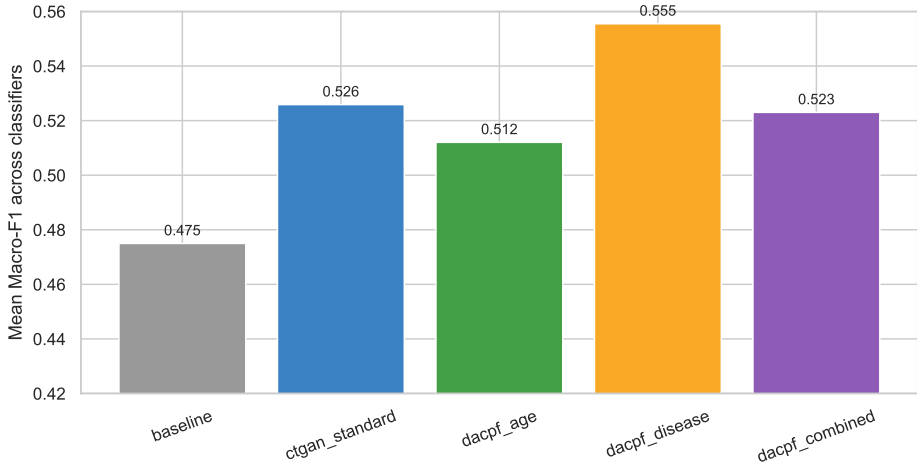


Figure 2: COVID-19 DACPF Macro-F1 averaged across CatBoost and XGBoost. The classifier-specific values in Table 3 show why this average should not be interpreted as a universal anchor effect.

This stage ruled out a simple monotonic story in which balancing classes, generating more

synthetic cases, or imposing more domain constraints necessarily improves prediction. We then used explainability and neighbourhood analyses to examine diversity, class-wise recall, feature-importance shifts, silhouette, nearest-neighbour label agreement, and hypothesis margin. Because some of these analyses used representative folds or earlier configurations, they establish the diagnostic rationale for what follows rather than a further confirmatory result. Their implication was that weak or unstable class geometry remained after augmentation, which led to the working hypothesis that class overlap, rather than imbalance alone, was limiting synthetic-to-real transfer.

## 4.2 Separability as an explicit generation objective

SEPAWARE is the intervention implied by that diagnosis. It does not change the target class counts; instead it asks whether records selected for label-consistent geometry transfer better to held-out real cases. The fixed- $\lambda$  experiment fits the complete pipeline independently in five stratified folds. Table 4 reports classifier-averaged fold means and 95%  $t$  intervals. On COVID-19, the balanced setting reaches 0.5492 Macro-F1, but its interval is wide. On Vigitel, both the varied grid and the fixed high-separability setting reach approximately 0.58 and outperform ordinary CTGAN in all ten matched fold-classifier comparisons. The varied-grid result remains exploratory because the configuration was selected using the same folds used for reporting.

Table 4: Five-fold TSTR Macro-F1. Means and 95% intervals are computed from five classifier-averaged fold scores.

| Dataset | Training data             | Mean          | 95% half-width |
|---------|---------------------------|---------------|----------------|
| COVID   | Real                      | 0.5069        | 0.0690         |
|         | Standard CTGAN            | 0.4627        | 0.0407         |
|         | SEPAWARE varied grid      | 0.5350        | 0.0649         |
|         | Fixed balanced (1.0, 0.5) | <b>0.5492</b> | 0.0959         |
| Vigitel | Real                      | 0.4672        | 0.0088         |
|         | Standard CTGAN            | 0.4588        | 0.0107         |
|         | SEPAWARE varied grid      | <b>0.5839</b> | 0.0225         |
|         | Fixed balanced (1.0, 0.5) | 0.5834        | 0.0216         |

Table 5 reports the highest mean selection scores from the same run. The repository composite fidelity score varies little across configurations, whereas SI and HM respond strongly to separability pressure. Figure 4 shows the same pattern across the full grid: global fidelity is comparatively stable while separability changes substantially. This is evidence that the selector controls a property not captured by the composite fidelity score; it is not evidence that the selected records are clinically representative.

## 4.3 Attributing the gain: replicated factorial experiment

The fixed- $\lambda$  results raise a sharper question: was the apparent gain produced by separability pressure, anchor pressure, or their interaction? The replicated  $2^2$  factorial experiment (Section 3.8) answers this component-level question under four predeclared settings. Figure 5 displays all 40 fold responses. COVID-19 profiles cross and show substantial fold-to-fold variation; Vigitel shows a stable step increase when separability is enabled.

The condition means in Table 6 make the contrast explicit. In Vigitel, enabling separability raises the mean from approximately 0.47–0.48 to 0.576–0.579, while adding anchors makes little

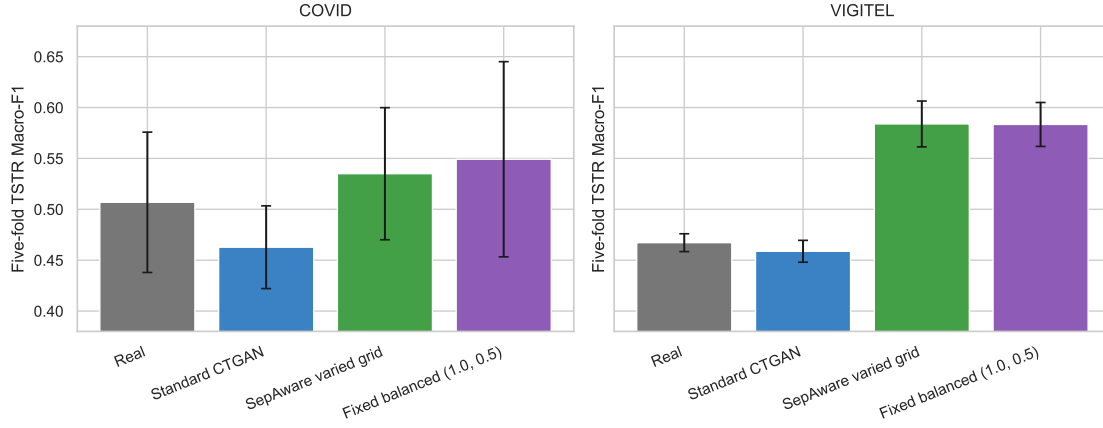


Figure 3: Five-fold TSTR Macro-F1 for the principal fixed- $\lambda$  comparison. Error bars are 95%  $t$  intervals over the five classifier-averaged fold scores.

Table 5: Top executed SEPARE grid configurations. “Quality” is the repository composite fidelity score.

| Dataset | $(\lambda_{sep}, \lambda_{anchor})$ | Quality | SI    | HM    | Selection |
|---------|-------------------------------------|---------|-------|-------|-----------|
| COVID   | (1.0, 0.0)                          | 0.956   | 0.779 | 0.897 | 2.154     |
| COVID   | (1.0, 0.25)                         | 0.961   | 0.760 | 0.854 | 2.095     |
| COVID   | (1.0, 0.5)                          | 0.965   | 0.754 | 0.830 | 2.067     |
| Vigitel | (1.0, 0.5)                          | 0.967   | 0.812 | 0.829 | 2.124     |
| Vigitel | (1.0, 0.25)                         | 0.967   | 0.811 | 0.822 | 2.116     |
| Vigitel | (1.0, 0.0)                          | 0.966   | 0.811 | 0.822 | 2.116     |

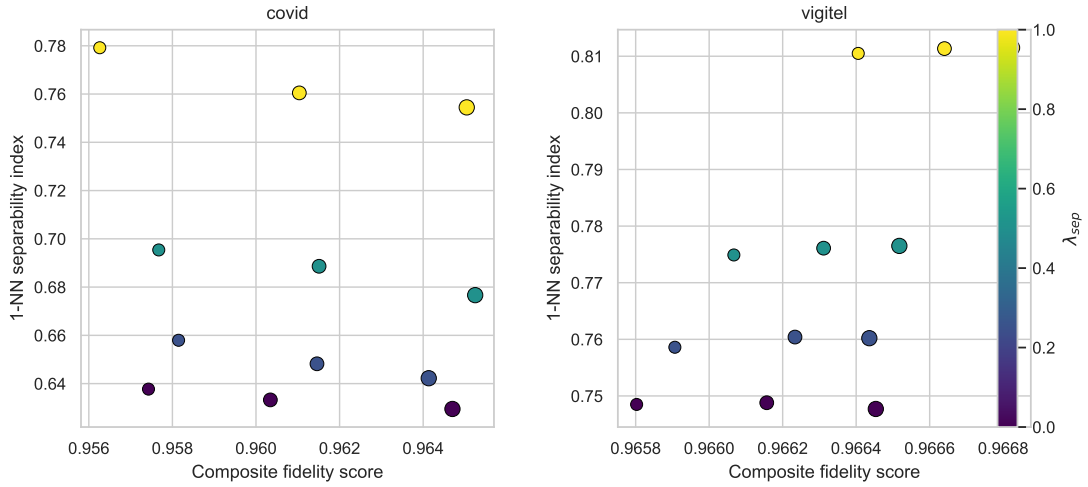


Figure 4: Five-fold mean fidelity–separability trade-off across executed  $\lambda$  settings. Colour represents separability weight; marker size represents anchor weight.

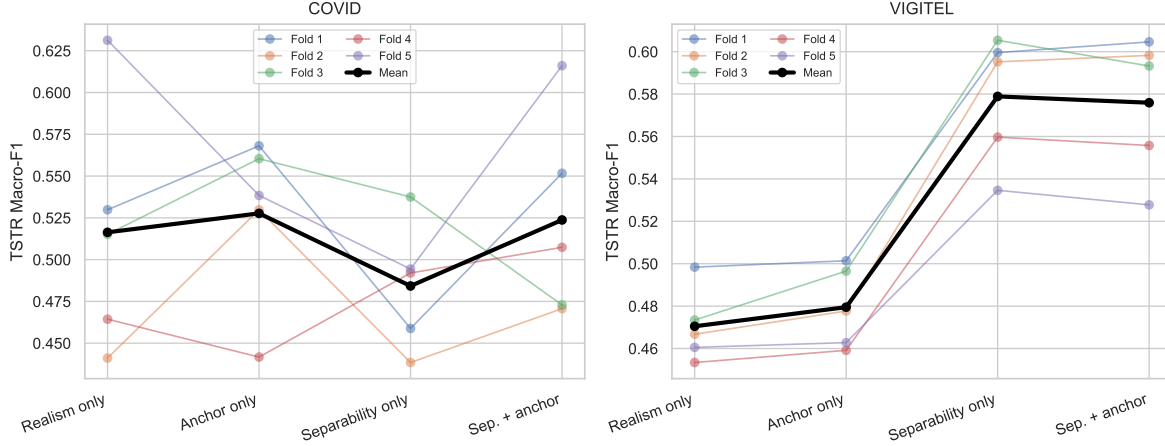


Figure 5: Fold-level TSTR Macro-F1 in the replicated  $2^2$  SEPAWARE experiment. Black lines show condition means.

Table 6: Mean TSTR Macro-F1 by predeclared factorial condition. Values are means of the five exported fold responses.

| Dataset | Realism only | Anchor only | Separability only | Both  |
|---------|--------------|-------------|-------------------|-------|
| COVID   | 0.516        | 0.528       | 0.484             | 0.524 |
| Vigitel | 0.470        | 0.480       | 0.579             | 0.576 |

difference. COVID-19 shows no corresponding ordering: separability-only is the *lowest* condition mean, and the combined condition returns approximately to realism-only performance.

The blocked raw-scale ANOVA in Table 7 is the primary component analysis. For COVID-19, none of the treatment effects is significant, and residual error accounts for 81.53% of treatment-plus-error variation after separating out the fold block. For Vigitel, separability accounts for 96.00% of that variation and is highly significant ( $p = 5.03 \times 10^{-10}$ ), while anchor and interaction contributions are negligible. The log-scale model yields the same substantive conclusion.

Table 7: Blocked factorial ANOVA: percentage of experimental variation and  $p$ -values on raw Macro-F1. Percentages exclude fold sum of squares from the denominator.

| Dataset | Source       | Variation (%) | $p$                    |
|---------|--------------|---------------|------------------------|
| COVID   | Separability | 5.13          | 0.4019                 |
|         | Anchor       | 10.21         | 0.2439                 |
|         | Interaction  | 3.13          | 0.5103                 |
|         | Error        | 81.53         | —                      |
| Vigitel | Separability | 96.00         | $5.03 \times 10^{-10}$ |
|         | Anchor       | 0.08          | 0.6069                 |
|         | Interaction  | 0.33          | 0.3154                 |
|         | Error        | 3.59          | —                      |

Normality tests do not reject for either dataset. Vigitel’s groupwise robust variance tests (Brown–Forsythe, Fligner–Killeen) do not reject equal variance, but Breusch–Pagan does, and the

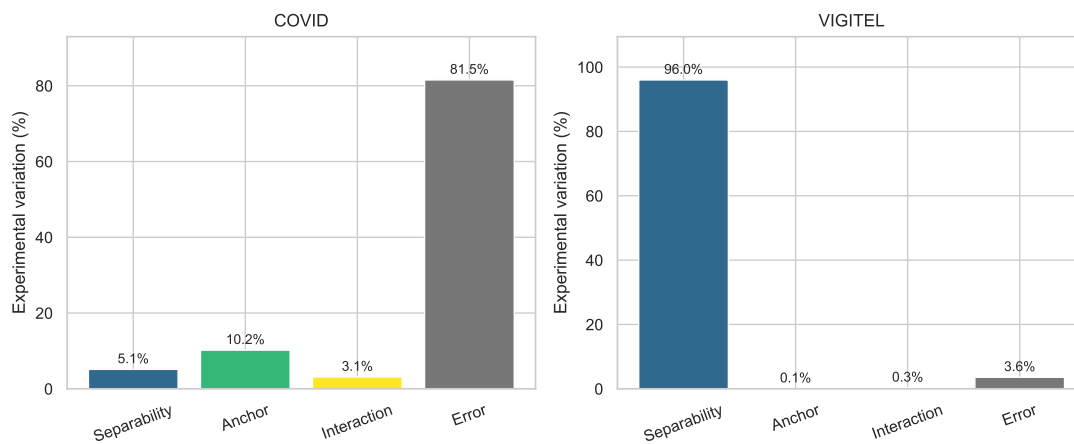


Figure 6: Variation allocation for the blocked raw-scale factorial models.

Table 8: Residual assumption diagnostics for the blocked factorial models.

| Dataset | Scale | Shapiro | B-F   | Fligner | B-P   | Assessment            |
|---------|-------|---------|-------|---------|-------|-----------------------|
| COVID   | raw   | .2446   | .8409 | .8394   | .4415 | no detected violation |
| COVID   | log   | .2031   | .8173 | .8050   | .6435 | no detected violation |
| Vigitel | raw   | .8564   | .8413 | .7706   | .0186 | variance flag         |
| Vigitel | log   | .8625   | .7463 | .6834   | .0237 | variance flag         |

log transform does not resolve this conflict. More importantly, independence is not established, because the cross-validation training sets overlap and the four conditions within each fold share a candidate pool. The very small Vigitel residual should therefore be confirmed across independent generator and partition seeds before it is treated as a stable population effect (Section 5.6).

#### 4.4 Strengthened augmentation comparison across four tasks

The practical augmentation analysis gives a more restrained result than the synthetic-to-real factorial experiment (Table 9). Across three repeated partitions and three classifiers, SEPAWARE CTGAN produced the highest COVID-19 Macro-F1 (0.541), followed by random oversampling (0.535) and class weighting (0.533). Random oversampling produced the highest Vigitel Macro-F1 (0.575), approximately equal to class weighting (0.574). Vigitel AUPRC was highest for real-only training (0.288) and class weighting (0.287); COVID-19 AUPRC was similar for random oversampling (0.357), SEPAWARE CTGAN (0.354), and real-only training (0.354).

Within-generator paired comparisons are favorable for CTGAN on three tasks but not the fourth. Relative to standard CTGAN, SEPAWARE increases mean AUPRC by 0.015 on COVID-19, 0.013 on Vigitel obesity, and 0.037 on Vigitel smoking; corresponding Macro-F1 differences are 0.033, 0.019, and 0.038. Kidney mortality is effectively unchanged (AUPRC difference 0.0003; Macro-F1 difference  $-0.002$ ). TVAE differences are inconsistent:  $-0.015$ ,  $-0.001$ ,  $0.007$ , and  $-0.0006$  AUPRC for COVID-19, Vigitel obesity, Vigitel smoking, and kidney mortality, respectively.

For Vigitel smoking, real-only training gives the highest AUPRC (0.299), whereas random oversampling gives the highest Macro-F1 (0.582). For kidney mortality, real-only training gives the highest AUPRC (0.121), followed by random undersampling (0.118). Real-only kidney models have zero mean minority recall at the default 0.5 threshold, while random undersampling raises recall to 0.620 but worsens the Brier score from 0.050 to 0.225. Thus ranking, thresholded sensitivity, and calibration lead to different operational conclusions.

Table 9: Held-out-real augmentation performance for the two new tasks, averaged across 3 repeated partitions, 5 folds, and 3 classifiers. AUPRC was the prespecified primary endpoint.

| Dataset          | Method               | AUPRC | Macro-F1 | Minority F1 | Brier |
|------------------|----------------------|-------|----------|-------------|-------|
| Vigitel smoking  | Real only            | 0.299 | 0.497    | 0.060       | 0.099 |
|                  | Random oversampling  | 0.290 | 0.582    | 0.313       | 0.165 |
|                  | Random undersampling | 0.283 | 0.558    | 0.331       | 0.210 |
|                  | SMOTE                | 0.232 | 0.540    | 0.186       | 0.134 |
|                  | Class weighting      | 0.295 | 0.574    | 0.298       | 0.164 |
|                  | Standard CTGAN       | 0.221 | 0.539    | 0.200       | 0.148 |
|                  | SepAware CTGAN       | 0.258 | 0.577    | 0.268       | 0.140 |
|                  | Standard TVAE        | 0.252 | 0.525    | 0.128       | 0.111 |
|                  | SepAware TVAE        | 0.259 | 0.529    | 0.134       | 0.110 |
| Kidney mortality | Real only            | 0.121 | 0.486    | 0.000       | 0.050 |
|                  | Random oversampling  | 0.107 | 0.529    | 0.166       | 0.156 |
|                  | Random undersampling | 0.118 | 0.490    | 0.174       | 0.225 |
|                  | SMOTE                | 0.095 | 0.521    | 0.128       | 0.132 |
|                  | Class weighting      | 0.108 | 0.525    | 0.162       | 0.157 |
|                  | Standard CTGAN       | 0.111 | 0.528    | 0.118       | 0.094 |
|                  | SepAware CTGAN       | 0.112 | 0.526    | 0.117       | 0.097 |
|                  | Standard TVAE        | 0.098 | 0.498    | 0.031       | 0.060 |
|                  | SepAware TVAE        | 0.097 | 0.499    | 0.034       | 0.062 |

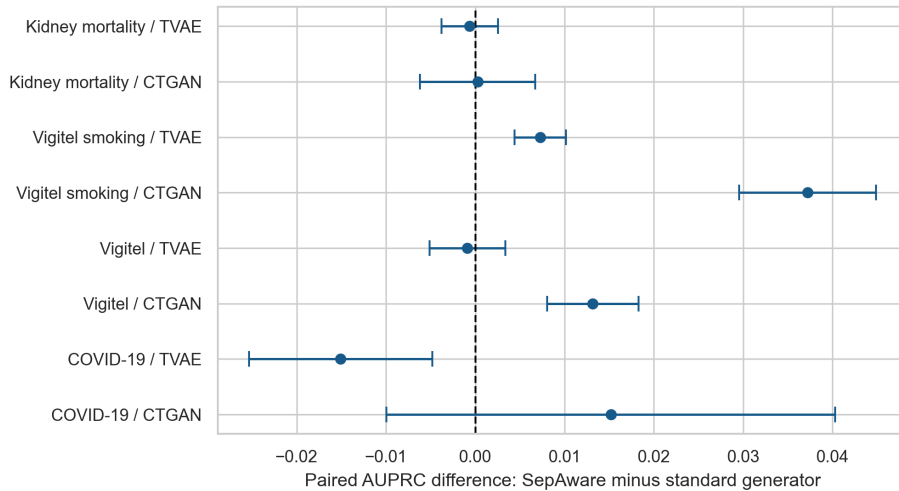


Figure 7: Paired AUPRC differences between SEPARE and the corresponding standard generator across shared fold-classifier observations. Error bars are conventional 95%  $t$  intervals and are descriptive because cross-validation folds are not independent replications.

The structural diagnostics reveal the trade-off behind selection. For Vigitel smoking, SEPARE CTGAN improves minority coverage distance from 3.81 to 3.62 but reduces diversity from 7.36 to 6.13. For kidney mortality, coverage improves from 1.74 to 1.65 while diversity falls from 6.91 to 4.09, without meaningful AUPRC improvement. Membership-proxy AUROC remains close to

chance. Exact standardized matches are absent for smoking and rare for kidney mortality (0.006% for SEPAWARE CTGAN), but these diagnostics do not establish anonymity.

These results do not support the claim that increasing synthetic separability is generally better. They support a narrower conclusion: within CTGAN candidate pools, separability-aware selection can yield a small average improvement and closer minority coverage, but the gain may come with reduced diversity and can be dominated by simpler frequency-correction methods.

## 4.5 Controlled prevalence and overlap

The controlled experiment confirms that prevalence and overlap exert distinct effects on rare-outcome performance. AUPRC increases sharply as separation rises and as the positive class becomes more prevalent. For logistic regression, SepAware-SMOTE exceeds ordinary SMOTE in all 9 prevalence–overlap cells, but the mean differences are small (0.001–0.024). The largest gains occur at 5% prevalence with moderate or low overlap and at 15% prevalence with high or moderate overlap; the benefit approaches zero at 30% prevalence with low overlap (Figure 8).

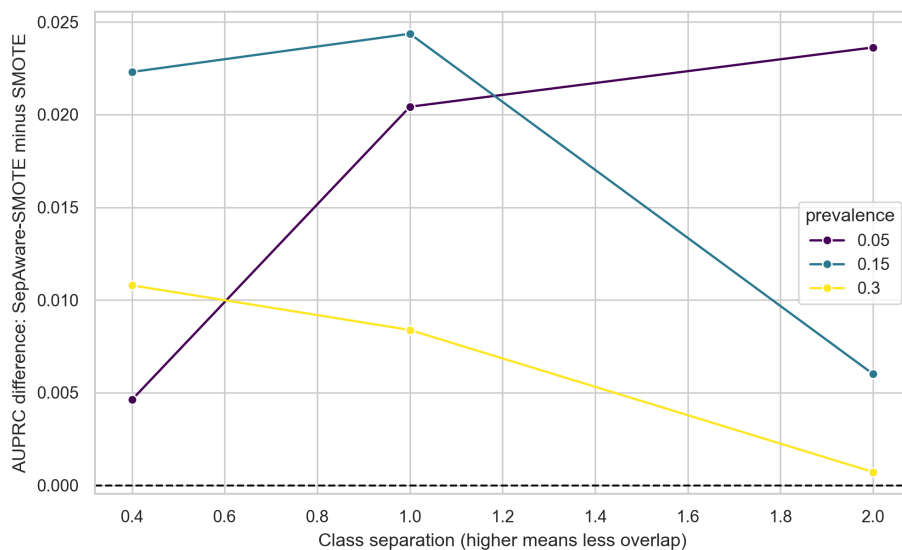


Figure 8: Controlled mechanism experiment for logistic regression. Each point is the mean AUPRC difference between SepAware-selected and ordinary SMOTE across 10 independent datasets. Higher class separation indicates less overlap.

In the pooled robust regression including both classifiers, neither the SepAware-SMOTE main contrast nor its interactions with overlap or prevalence were statistically distinguishable from zero (all  $P > .91$ ). The simulation therefore supports the distinction between prevalence and overlap but does not establish a stable region in which separability-aware SMOTE selection is superior. It provides no evidence for imposing the hypothesized inverted-U relationship on the observed health datasets.

## 4.6 Summary of the empirical evidence

The experiments form a coherent progression. Rebalancing and unconstrained augmentation did not make utility reliably high; domain filters sometimes helped but did not behave monotonically; diagnostic analyses implicated class overlap; SEPAWARE then improved Vigitel TSTR performance

while measured composite fidelity remained broadly stable; and the factorial study attributed that gain to separability rather than to anchors. The COVID-19 null result is equally informative: explicit separability pressure is not sufficient when the positive class is small, heterogeneous, or weakly represented.

The strengthened augmentation results qualify that progression. CTGAN selection improves average AUPRC on COVID-19 and both Vigitel tasks but not kidney mortality; random oversampling remains highly competitive, and TVAE selection is inconsistent. Coverage gains repeatedly accompany lower minority diversity. The evidence therefore does not show that SEPAWARE universally recovers clinically meaningful signal; it may instead concentrate easier observations.

## 5 Discussion

### 5.1 Principal findings

The study produces two complementary findings. First, the synthetic-to-real factorial experiment shows that separability is an attributable, utility-relevant property in Vigitel obesity, whereas COVID-19 attribution is not established because residual fold variation dominates. Second, the four-task augmentation analysis shows that this substitution result is not a general augmentation advantage. SEPAWARE improves CTGAN relative to standard CTGAN for COVID-19 and both Vigitel outcomes, but not kidney mortality; random oversampling is competitive and TVAE results are inconsistent.

This distinction matters. TSTR asks whether a synthetic dataset can substitute for real training data; augmentation asks whether synthetic minority records add value when the original training records remain available. A method may improve the former without being the best practical solution for the latter. For Vigitel, real-only AUPRC was 0.288, random oversampling achieved 0.284, and SEPAWARE CTGAN achieved 0.250. Its advantage was relative to standard CTGAN, not relative to every alternative. For COVID-19, random oversampling achieved the highest mean AUPRC (0.357), while SEPAWARE CTGAN and real-only training both achieved 0.354.

### 5.2 Comparison with prior work

The competitiveness of random oversampling agrees with a 2026 JMIR study in which sampling with replacement matched substantially more complex augmentation in small-data settings [18]. Our results extend that caution to rare outcomes: greater model complexity does not ensure that augmentation adds information. The generator and its sampling mechanism can matter as much as the downstream selector.

Contemporary evaluation frameworks emphasize that fidelity, utility, and privacy can move in different directions [15, 16]. We observe the same pattern within the minority class. SEPAWARE CTGAN improves nearest-minority coverage while reducing pairwise minority diversity. A single aggregate fidelity score would not expose that trade-off. The finding also supports calls for explicit privacy and utility threat models rather than duplicate checks alone [3].

### 5.3 When separability helps

The four-task contrast suggests that separability pressure interacts with outcome structure and candidate-pool quality. Vigitel smoking resembles Vigitel obesity in showing a CTGAN-selection gain, although neither beats the strongest simple baseline on AUPRC. Kidney mortality behaves

differently: selection improves geometric coverage and reduces diversity without adding ranking utility. This is direct evidence against a universal “more separability is better” interpretation.

The negative TVAE result is equally important. SDV’s TVAE does not support native outcome-conditional sampling, so the experiment fitted a minority-conditional feature model. The resulting synthetic minorities had very low diversity, especially for Vigitel. Post-selection could not recover information absent from the candidate pool and sometimes reduced performance further. SEPAWARE is therefore not generator agnostic merely because its scoring rule can technically be applied to another generator.

## 5.4 The easier-case explanation

Higher separability can represent clearer class-conditional signal, but it can also represent deletion of difficult cases. The strengthened diagnostics support both interpretations. Smaller held-out-minority coverage distance is favorable: selected CTGAN records lie closer to real positive test cases. Lower within-minority diversity is unfavorable: the selector concentrates the candidate distribution. Because no externally validated phenotype labels are available, we cannot determine whether the lost variation is noise, redundant spread, or clinically meaningful heterogeneity.

This trade-off changes the decision rule for the method. A Macro-F1 or AUPRC gain alone is insufficient. A favorable use case requires improvement relative to the corresponding standard generator without material loss of calibration, minority coverage, or clinically meaningful diversity. Boundary-stratified error analysis and expert-defined minority phenotypes are needed to determine how much separability is too much.

## 5.5 Privacy implications

No exact matches were observed and the membership proxy remained near chance. These are reassuring diagnostics but not evidence of anonymity. SEPAWARE explicitly prefers records near the real manifold, so the same mechanism that improves utility could increase disclosure risk under a stronger attacker. Membership and attribute inference under explicit attacker knowledge, rarity-stratified analyses, and formal privacy mechanisms remain necessary before any synthetic records are released.

## 5.6 Statistical interpretation

The factorial conditions share candidate pools within folds and cross-validation training sets overlap. The strengthened augmentation analysis uses three repeated five-fold partitions, two generators, three classifiers, and four tasks, but its 45 fold-classifier observations per task remain dependent and the two Vigitel outcomes share one source. The reported paired intervals are therefore descriptive rather than independent clinical replications.

The appropriate conclusion is conditional. Separability-aware selection controls a property not captured by class balance or global fidelity and can improve some CTGAN pools. It is not uniformly beneficial across outcomes or generators, does not consistently beat simple oversampling, and can purchase coverage gains with reduced diversity. Those boundary conditions are the scientific contribution of the four-task study.

## 6 Limitations

Four outcome tasks from three health-data sources were evaluated. This broadens the observed prevalence and outcome contexts but does not identify a general relationship between intrinsic overlap and SepAware benefit; the two Vigitel tasks are not independent sources. The study should therefore be read as an initial multitask validation with a controlled mechanism experiment.

The strongest Vigitel factorial finding comes from one 5000-record sample, one cross-validation partition, one CTGAN configuration, and shared within-fold candidate pools. The strengthened experiment uses three repeated five-fold partitions and 30 training epochs for computational feasibility. Fold-specific generator seeds avoid reusing a single fitted generator, and repeated shuffles measure some partition variability, but all repetitions still reuse the same two source samples. More generator seeds, higher-capacity sensitivity runs, and independent datasets are necessary before formal confirmatory claims.

The fixed lambda grid was exploratory because performance and configuration selection used the same folds. The strengthened analysis fixes the separability weight in advance, but it does not estimate a clinically optimal trade-off between discrimination and minority diversity. The score also depends on a random-forest confidence term; additional boundary models and label-permutation controls are needed to test classifier-specific shortcuts.

The TVAE condition is not directly symmetric with CTGAN. SDV 1.17 does not implement native TVAE conditional sampling, so TVAE was fitted to minority-class features. This is a reproducible conditional-generation strategy, but its low diversity shows that post-selection cannot compensate for a collapsed or highly concentrated candidate pool. A diffusion model and a generator with native conditional support should be added before making a broad generator-agnostic claim.

Minority coverage and diversity are geometric proxies rather than clinically validated phenotype measures. The COVID-19 death class may contain several pathways that are not represented by unsupervised distances. Blinded clinician review and prespecified phenotype or subgroup definitions are required to determine whether removed variation is clinically meaningful. Calibration estimates are also unstable in the small COVID-19 folds.

The privacy analysis is limited to exact matches, nearest-record distances, and a membership proxy. It does not include attribute inference, shadow-model membership attacks, linkage attacks, or a formal differential-privacy guarantee. No synthetic data release should be justified from the present privacy results.

Finally, external or temporal validation is absent. The smoking file contains no survey-year field, the obesity evaluation is not a prospective temporal split, and the COVID-19 cohort is institution-specific. Kidney mortality uses a longitudinal landmark but random patient partitions rather than a calendar-time split. At least one temporal or cross-site evaluation is needed.

These limitations define the next additions: a native conditional third generator, multiple generator seeds per fold, expert-informed minority coverage, stronger privacy attacks, formal overlap-to-benefit modeling, and external or temporal validation. Failures should continue to be reported as operational guidance about when separability control should not be used.

## 7 Conclusions

Class imbalance and class overlap are not interchangeable problems. In the synthetic-to-real factorial experiment, explicit separability pressure explains the large Vigitel-obesity utility change but not the COVID-19 variation. In four-task augmentation, SepAware improves CTGAN for COVID-

19 and both Vigitel outcomes but not kidney mortality, is inconsistent for TVAE, and is often matched or exceeded by simple rebalancing. Closer minority coverage is accompanied by reduced diversity.

SepAware should therefore be understood as an auditable mechanism for controlling and studying class-conditional geometry, not as a universally superior augmentation method. Its practical use requires evidence that a predictive gain survives simple baselines and does not come from removing difficult but legitimate minority cases. The four-task study establishes this conditional direction and identifies kidney mortality as a clear nonbeneficial boundary case.

## Abbreviations

**AUPRC** area under the precision-recall curve

**AUROC** area under the receiver operating characteristic curve

**CTGAN** conditional tabular generative adversarial network

**HM** hypothesis margin

**SI** separability index

**SMOTE** synthetic minority oversampling technique

**TSTR** train on synthetic, test on real

**TVAE** tabular variational autoencoder

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## A Reproducibility and Artefact Status

Table 10 lists the status of every research artefact consulted while preparing this paper and states how each was treated. Superseded, protocol-only, or failed runs are excluded from all numerical claims in the main text; only executed, leakage-controlled five-fold notebooks and their exported CSV tables are reported as evidence.

Table 10: Status of research artefacts consulted during manuscript preparation.

| Artefact                                                  | Status                            | Manuscript treatment                                                                                                                                    |
|-----------------------------------------------------------|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Older dataset-specific fixed- $\lambda$ notebooks         | Superseded single-split studies   | Excluded from numerical claims because they use one 75/25 split.                                                                                        |
| Combined SEPAWARE fixed- $\lambda$ notebook               | Executed five-fold study          | Authority for fold-level fixed- $\lambda$ comparisons (Table 4).                                                                                        |
| Protocol-audited 2 <sup>2</sup> cross-validation notebook | Executed                          | Superseded by the replicated factorial notebook below; retained only for earlier condition summaries and correlation interpretation during development. |
| Replicated factorial ANOVA notebook (with results)        | Latest executed experiment        | Authority for the 40 fold responses, ANOVA, effects, and diagnostics (Section 4.3).                                                                     |
| Factorial assumption-test script and CSVs                 | Executed after the notebook above | Authority for expanded residual tests and the blocked-versus-unblocked sensitivity analysis.                                                            |
| COVID DACPF raw and summary CSVs                          | Executed                          | Authority for augmentation results in Table 3.                                                                                                          |
| TabPFN augmentation notebook                              | Failed dependency, no output      | Protocol only; no result claimed.                                                                                                                       |
| Unexecuted causal-filter notebooks                        | Incomplete protocol               | Discussed as development history / future work, not evidence.                                                                                           |

| Artefact                                | Status                                     | Manuscript treatment                                                                                                                      |
|-----------------------------------------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Cluster-then-augment notebook           | No authoritative exported result located   | Excluded from numerical results.                                                                                                          |
| Four-task augmentation script           | Executed three-repeat five-fold experiment | Authority for conventional baselines, CTGAN/TVAE augmentation, calibration, positive-class coverage/diversity, and privacy-proxy results. |
| Controlled prevalence-by-overlap script | Executed mechanism experiment              | Authority for the independent prevalence and overlap manipulation; not counted as a third health dataset.                                 |

### A.1 Result-authority decisions

The rerun factorial experiment differs numerically from earlier protocol-audited notes because CTGAN is stochastic even under a fixed, recorded software stack; this paper uses the newly executed observations consistently and does not mix values across reruns. An unblocked  $2^2r$  decomposition that treats folds as replications is also present in the repository and assigns COVID-19 90.01% error and Vigitel 16.92% error; the main text instead uses the blocked model (Equation 2), which separates fold variation from treatment variation and yields 81.53% and 3.59% respectively (Table 7). Both calculations exist in the repository, but they answer different questions, and only the blocked model is used for the paper’s conclusions.

The qualification factorial protocol evaluates standalone synthetic datasets (TSTR). The strengthened experiment separately reports real-plus-synthetic augmentation and does not pool the two regimes. AUPRC, calibration, minority geometry, exact-match, and membership-proxy measures are reported only for the strengthened run. Attribute inference, formal privacy guarantees, and external validation remain absent and are reported as limitations rather than estimated after the fact.